MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 22: Friday 30 October 2026

(Week 11)

Remark. Q21.3 to Q21.5 are leftover exercises from the Tuesday class.

Remark. Q22.1 is additional exercise on triple integrals in spherical coordinates (Section 14.7).



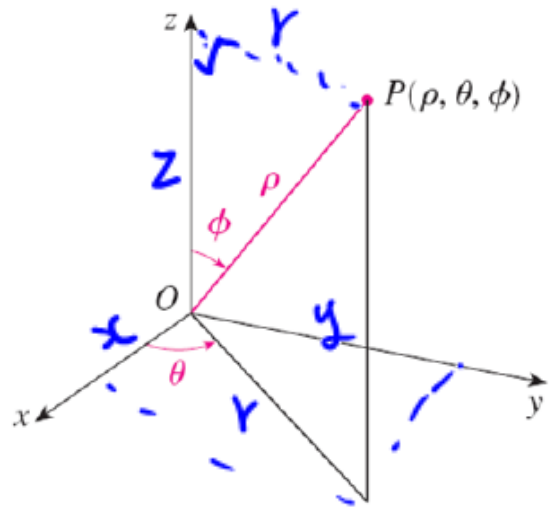
Recall: Triple integrals in spherical coordinates

Spherical coordinates (ρ, ϕ, θ):

- ρ is the distance from ***P*** to the origin ***O*** **($\rho \geq 0$)**.
- ϕ is the angle $\overrightarrow{OP}$ makes with the positive ***z***-axis. **($0 \leq \phi \leq \pi$)**
- θ is the same angle as in cylindrical coordinates.

cylindrical coordinates (r, θ, z)

$$\begin{cases} r = \rho \sin\phi \\ \theta = \theta \\ z = \rho \cos\phi \end{cases}$$



rectangular coordinates (x, y, z)

- $x = \rho \sin\phi \cos\theta$
- $y = \rho \sin\phi \sin\theta$
- $z = \rho \cos\phi$

Also, $\rho = \sqrt{x^2+y^2+z^2} = \sqrt{r^2+z^2}$.

Transformation between (x, y, z) and (ρ, ϕ, θ):

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho \cos\phi. \qquad (1)$$

Suppose R is a region in xyz-space, and G is the corresponding region in the $\rho\phi\theta$-space. Then

$$\iiint_R F(x, y, z)\, dV \qquad (2)$$

$$= \iiint_G F(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)\, \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

Q21.3. Evaluate the iterated integral

$$\int_{\pi/6}^{\pi/3} \int_{\csc\phi}^{2\csc\phi} \int_{0}^{2\pi} \rho^2 \sin\phi \, d\theta \, d\rho \, d\phi.$$

Solution:

$$\int_{\pi/6}^{\pi/3} \int_{\csc\phi}^{2\csc\phi} \int_{0}^{2\pi} \rho^2 \sin\phi \, d\theta \, d\rho \, d\phi$$

$$= \int_{\pi/6}^{\pi/3} \int_{\csc\phi}^{2\csc\phi} \left(\int_{0}^{2\pi} \rho^2 \sin\phi \, d\theta \right) d\rho \, d\phi$$

$$= \int_{\pi/6}^{\pi/3} \int_{\csc\phi}^{2\csc\phi} \rho^2 \sin\phi \left(\int_{0}^{2\pi} 1 \, d\theta \right) d\rho \, d\phi$$

$$= \left(\int_{0}^{2\pi} 1 \, d\theta \right) \cdot \int_{\pi/6}^{\pi/3} \left(\int_{\csc\phi}^{2\csc\phi} \rho^2 \sin\phi \, d\rho \right) d\phi$$

$$= \left(\int_{0}^{2\pi} 1 \, d\theta \right) \cdot \int_{\pi/6}^{\pi/3} \sin\phi \left(\int_{\csc\phi}^{2\csc\phi} \rho^2 \, d\rho \right) d\phi$$

$$= \left[\theta \right]_{0}^{2\pi} \cdot \int_{\pi/6}^{\pi/3} \sin\phi \left[\frac{\rho^3}{3} \right]_{\csc\phi}^{2\csc\phi} d\phi$$

$$= (2\pi - 0) \cdot \int_{\pi/6}^{\pi/3} \sin\phi \left(\frac{8\csc^3\phi}{3} - \frac{\csc^3\phi}{3} \right) d\phi$$

$$= 2\pi \cdot \int_{\pi/6}^{\pi/3} \sin\phi \cdot \frac{7}{3} \csc^2\phi \, d\phi$$

$$= \frac{14\pi}{3} \int_{\pi/6}^{\pi/3} \csc^2\phi \, d\phi$$

$\frac{d}{d\phi}(\cot\phi) = -\csc^2\phi$

$$= \frac{14\pi}{3} \Big[-\cot\phi \Big]_{\pi/6}^{\pi/3}$$

$$= \frac{14\pi}{3} \left(-\cot\left(\frac{\pi}{3}\right) - \left(-\cot\left(\frac{\pi}{6}\right) \right) \right)$$

$$= \frac{14\pi}{3} \left(-\frac{1}{\sqrt{3}} + \sqrt{3} \right) = \frac{14\pi}{3} \cdot \frac{-1+3}{\sqrt{3}}$$

$$= \frac{14\pi}{3} \cdot \frac{2}{\sqrt{3}} = \frac{28\pi}{3\sqrt{3}} = \frac{28\sqrt{3}\pi}{9}.$$

Q21.4. Evaluate the following integral:

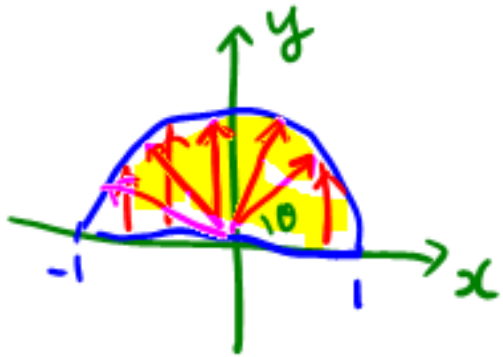
$$\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} \sqrt{x^2+y^2+z^2}\; dz\, dy\, dx.$$

$$= \iiint_E \sqrt{x^2+y^2+z^2}\, dV$$

$$E = \{(x,y,z) : -1 \le x \le 1,\ 0 \le y \le \sqrt{1-x^2},\ \sqrt{x^2+y^2} \le z \le \sqrt{2-x^2-y^2}\}$$

$$= \{(r,\theta,z) : 0 \le \theta \le \pi,\ 0 \le r \le 1,\ r \le z \le \sqrt{2-r^2}\}$$

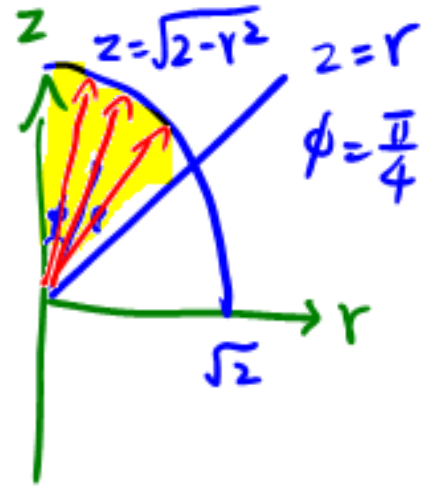
For a given θ,

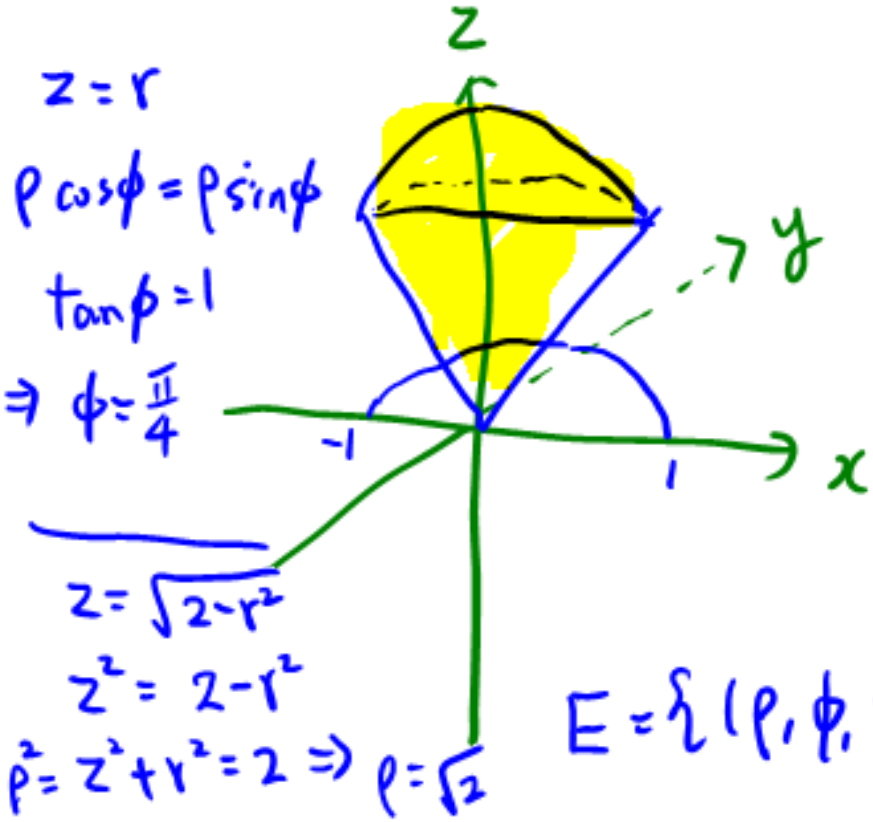


$y = \sqrt{1-x^2} \ge 0$

$y^2 = 1 - x^2$

$x^2 + y^2 = 1$





$z = r$

$\rho\cos\phi = \rho\sin\phi$

$\tan\phi = 1$

$\Rightarrow \phi = \frac{\pi}{4}$

$z = \sqrt{2-r^2}$

$z^2 = 2 - r^2$

$\rho^2 = z^2 + r^2 = 2 \Rightarrow \rho = \sqrt{2}$

$$\sqrt{x^2+y^2} \le z \le \sqrt{2-(x^2+y^2)}$$

$$\Leftrightarrow \quad r \le z \le \sqrt{2-r^2}$$

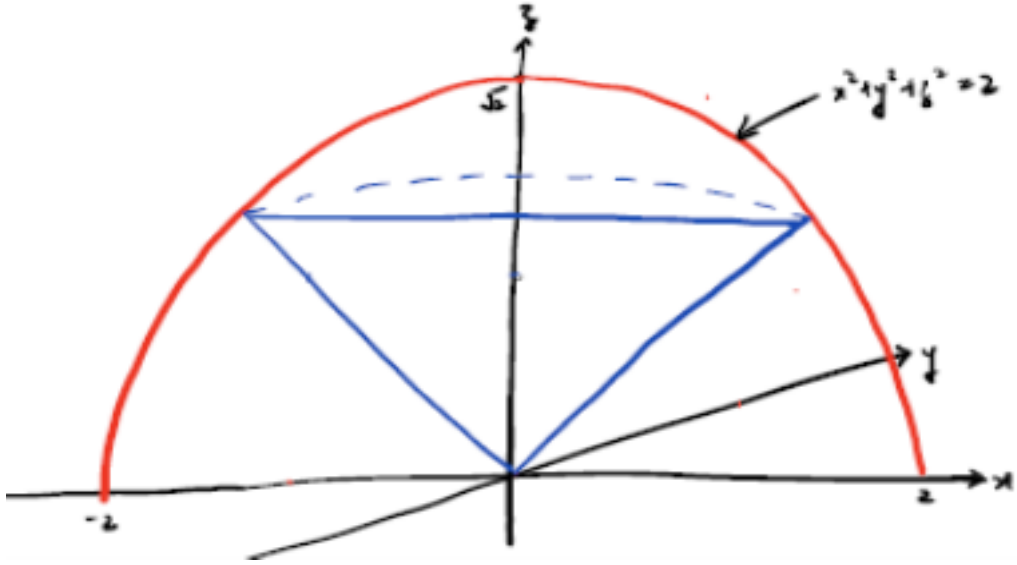
$z = \sqrt{2-r^2} \ge 0$

$z^2 = 2 - r^2$

$z^2 + r^2 = (\sqrt{2})^2$

$$E = \left\{(\rho,\phi,\theta) : 0 \le \theta \le \pi,\ 0 \le \phi \le \tfrac{\pi}{4},\ 0 \le \rho \le \sqrt{2}\right\}$$

Solution:



Let

$$E = \{(x, y, z) : -1 \le x \le 1,\ 0 \le y \le \sqrt{1 - x^2},$$
$$\sqrt{x^2 + y^2} \le z \le \sqrt{2 - x^2 - y^2}\}$$
$$= \{(r, \theta, z) : 0 \le \theta \le \pi,\ 0 \le r \le 1,\ r \le z \le \sqrt{2 - r^2}\}$$
$$E = \{(\rho, \theta, \phi) : 0 \le \theta \le \pi,\ 0 \le \phi \le \frac{\pi}{4},\ 0 \le \rho \le \sqrt{2}\}$$

Thus

$$\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}}\sqrt{x^2+y^2+z^2}\;dz\,dy\,dx$$

$$=\iiint_E \sqrt{x^2+y^2+z^2}\,dV \qquad \rho^2 = x^2+y^2+z^2$$

$$=\int_0^{\pi}\int_0^{\frac{\pi}{4}}\left(\int_0^{\sqrt{2}}\rho\cdot\rho^2\sin\phi\,d\rho\right)d\phi\,d\theta$$

$$=\int_0^{\pi}\int_0^{\frac{\pi}{4}}\sin\phi\left[\frac{\rho^4}{4}\right]_0^{\sqrt{2}}d\phi\,d\theta$$

$$=\int_0^{\pi}\left(\int_0^{\frac{\pi}{4}}\sin\phi\cdot(1-0)\,d\phi\right)d\theta$$

$$=\int_0^{\pi}\Big[-\cos\phi\Big]_0^{\frac{\pi}{4}}d\theta=\int_0^{\pi}\left(-\frac{\sqrt{2}}{2}-(-1)\right)d\theta$$

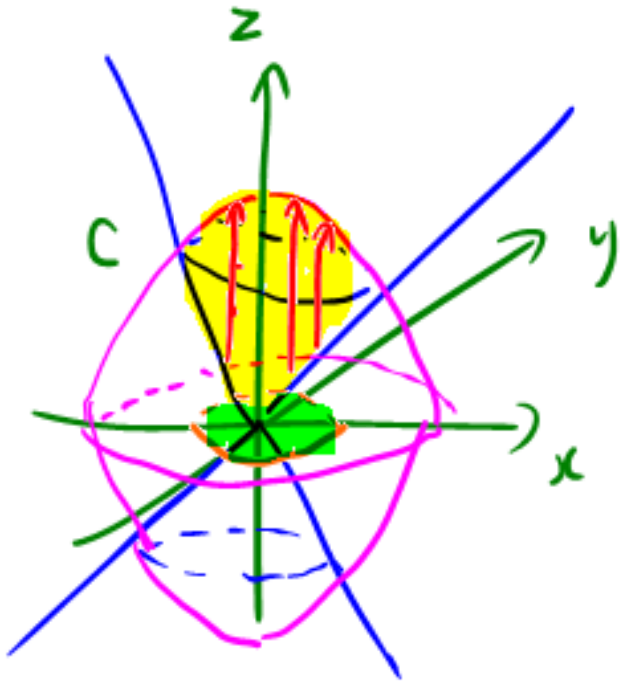
$$=\left(1-\frac{\sqrt{2}}{2}\right)\Big[\theta\Big]_0^{\pi}=\left(1-\frac{\sqrt{2}}{2}\right)\pi.$$

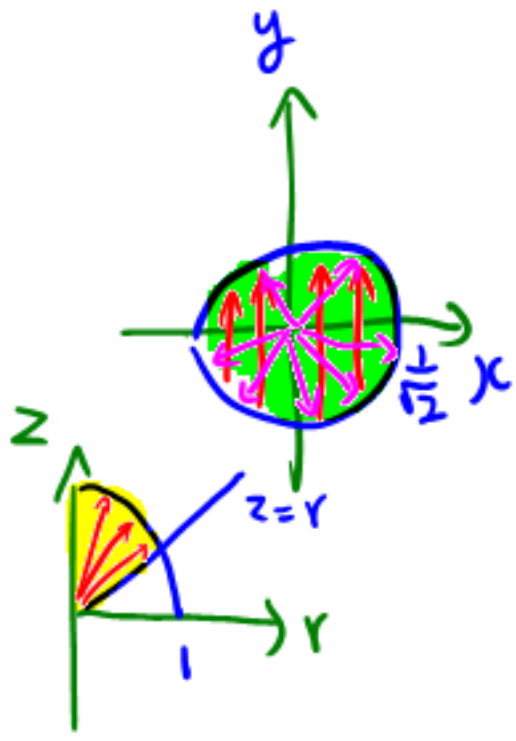
Q21.5. Determine the volume of the region D in the xyz-space bounded below by the cone $z^2 = x^2 + y^2$ and bounded above by the sphere of radius **1** centered at the origin.

$$z^2 = x^2 + y^2$$

$$\Leftrightarrow z = \pm\sqrt{x^2+y^2}$$

$$D = \left\{ (x,y,z) : -\frac{1}{\sqrt{2}} \le x \le \frac{1}{\sqrt{2}}, -\sqrt{\tfrac{1}{2}-x^2} \le y \le \sqrt{\tfrac{1}{2}-x^2}, \sqrt{x^2+y^2} \le z \le \sqrt{1-x^2-y^2} \right\}$$





$$\begin{cases} z^2 = x^2+y^2 \\ x^2+y^2+z^2 = 1 \end{cases}$$

$$\Rightarrow z^2 + z^2 = 1$$

$$\Rightarrow z^2 = \frac{1}{2}$$

$$\Rightarrow C: \begin{cases} z = \frac{1}{\sqrt{2}} \\ x^2+y^2 = \frac{1}{2} \end{cases}$$

$$D = \{(r,\theta,z): \ 0 \le \theta \le 2\pi, \ 0 \le r \le \tfrac{1}{\sqrt{2}}, \ r \le z \le \sqrt{1-r^2}\}$$

$$D = \{(\rho,\phi,\theta): \ 0 \le \theta \le 2\pi, \ 0 \le \phi \le \tfrac{\pi}{4}, \ 0 \le \rho \le 1\}$$

$$\sqrt{x^2+y^2} \le z \le \sqrt{1-x^2-y^2}$$

$$r \le z \le \sqrt{1-r^2}$$

$$z = \sqrt{1-r^2}$$

$$z^2 = 1-r^2 \Rightarrow r^2+z^2 = 1$$

$$\text{Volume}(D) = \iiint_D 1 \, dV$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^1 1 \cdot \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

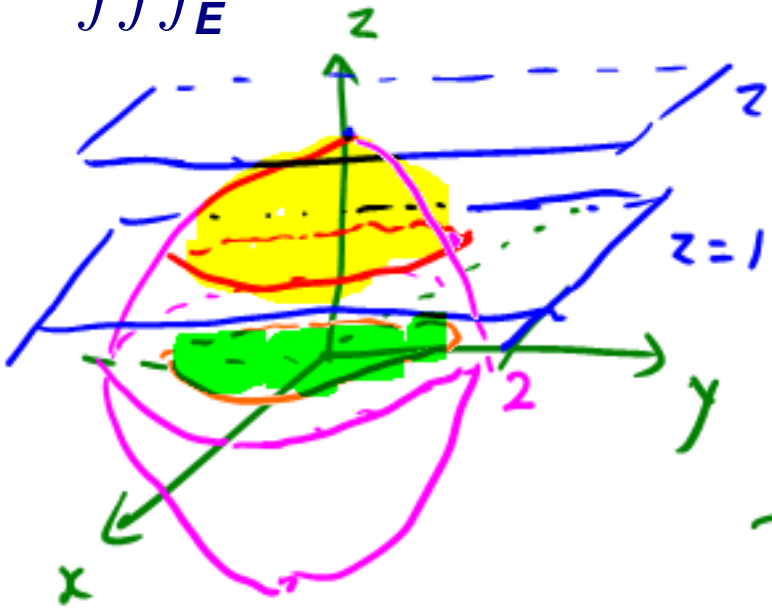
Solution:

$$
\begin{aligned}
V &= \int_0^{2\pi} \int_0^{\pi/4} \left(\int_0^1 \rho^2 \sin\phi \, d\rho \right) d\phi \, d\theta \\
&= \int_0^{2\pi} \int_0^{\pi/4} \left(\int_0^1 \rho^2 \sin\phi \, d\rho \right) d\phi \, d\theta \\
&= \int_0^{2\pi} \int_0^{\pi/4} \sin\phi \left(\int_0^1 \rho^2 \, d\rho \right) d\phi \, d\theta \\
&= \left(\int_0^1 \rho^2 \, d\rho \right) \cdot \int_0^{2\pi} \left(\int_0^{\pi/4} \sin\phi \, d\phi \right) d\theta \\
&= \left(\int_0^1 \rho^2 \, d\rho \right) \cdot \left(\int_0^{\pi/4} \sin\phi \, d\phi \right) \cdot \int_0^{2\pi} 1 \, d\theta \\
&= \left(\int_0^1 \rho^2 \, d\rho \right) \cdot \left(\int_0^{\pi/4} \sin\phi \, d\phi \right) \cdot \left(\int_0^{2\pi} d\theta \right)
\end{aligned}
$$

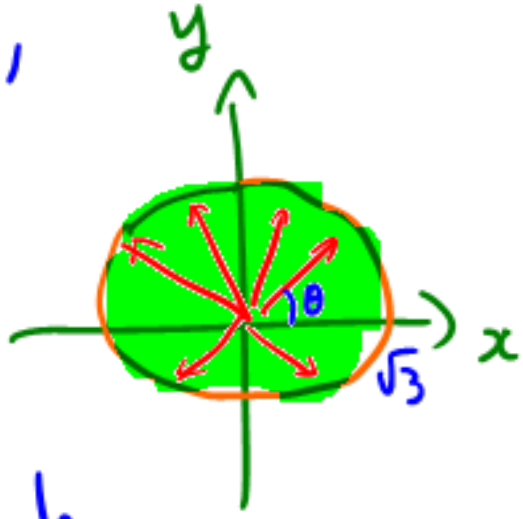
$$
\begin{aligned}
&= \left[\frac{\rho^3}{3}\right]_0^1 \cdot \Big[-\cos\phi\Big]_0^{\pi/4} \cdot \Big[\theta\Big]_0^{2\pi} \\
&= \left(\frac{1^3}{3} - \frac{0^3}{3}\right) \cdot \left(-\frac{\sqrt{2}}{2} - (-1)\right) \cdot (2\pi - 0) \\
&= \frac{1}{3} \cdot \left(1 - \frac{\sqrt{2}}{2}\right) \cdot 2\pi \\
&= \frac{2\pi}{3}\left(1 - \frac{\sqrt{2}}{2}\right) \\
&= \frac{\pi}{3}(2 - \sqrt{2}).
\end{aligned}
$$

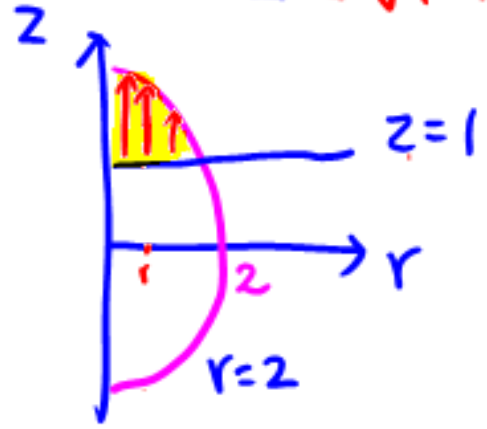
Q22.1. Consider the solid region E in the xyz-space consisting of points (x, y, z) such that $1 \leq z \leq 2$ and $x^2 + y^2 + z^2 \leq 4$. Use spherical coordinates to evaluate $\iiint_E (x^2 + y^2 + z^2)^{-1/2}\, dV$.

$x^2+y^2+z^2=4 \Rightarrow r^2+z^2=4$
$\Rightarrow z=\sqrt{4-r^2}$



$E=\{(r,\theta,z): 0\le\theta\le 2\pi,$
$0\le r\le\sqrt{3}$
$1\le z\le\sqrt{4-r^2}\}$



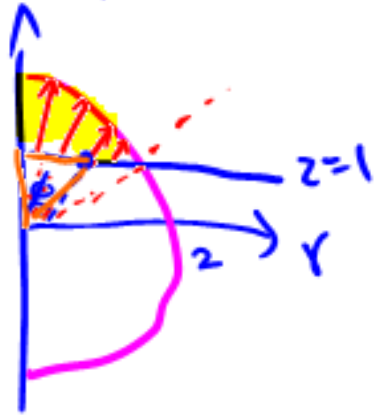


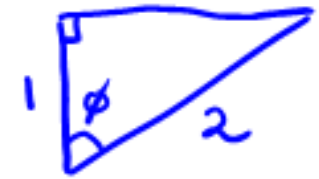
To get radius of the circle,

$$\begin{cases} z=1 \\ x^2+y^2+z^2=4 \end{cases} \Rightarrow x^2+y^2=4-1=3 \Rightarrow r=\sqrt{x^2+y^2}\Big|_{z=1}=\sqrt{3}$$

$$E = \{(\rho, \phi, \theta) : 0 \le \theta \le 2\pi,\ 0 \le \phi \le \tfrac{\pi}{3},\ \sec\phi \le \rho \le 2\}$$







$\cos\phi = \frac{1}{2} \Rightarrow \phi = \frac{\pi}{3}$

$\rho\cos\phi = 1$

$\rho = \sec\phi$

$$\iiint_E (x^2+y^2+z^2)^{-\frac{1}{2}}\, dV$$

$$= \int_0^{2\pi}\int_0^{\frac{\pi}{3}}\int_{\sec\phi}^{2} (\rho^2)^{-\frac{1}{2}}\, \rho^2\sin\phi\, d\rho\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{\pi}{3}}\left(\int_{\sec\phi}^{2} \rho\sin\phi\, d\rho\right) d\phi\, d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{3}} \sin\phi \left[\frac{\rho^2}{2}\right]_{\sec\phi}^{2} d\phi\, d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{3}} \sin\phi \cdot \left[2 - \frac{\sec^2\phi}{2}\right] d\phi\, d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{3}} \left(2\sin\phi - \frac{1}{2}\cdot\frac{\sin\phi}{\cos^2\phi}\right) d\phi\, d\theta$$

$$= \int_0^{2\pi} \left[-2\cos\phi - \frac{1}{2\cos\phi}\right]_0^{\frac{\pi}{3}} d\theta \qquad \cos\tfrac{\pi}{3} = \tfrac{1}{2}$$

$$= \int_0^{2\pi} \left[\left(-2\cdot\tfrac{1}{2} - \frac{1}{2\cdot\frac{1}{2}}\right) - \left(-2 - \tfrac{1}{2}\right)\right] d\theta$$

$$= \int_0^{2\pi} \frac{1}{2}\, d\theta = \left[\frac{\theta}{2}\right]_0^{2\pi} = \pi.$$